

June 2026

To the Editors,
Patterns, Cell Press
Attn: Dr. Alejandra Alvarado

Dear Dr. Alvarado,

Thank you for your assessment of our earlier submission (PATTERNS-D-26-00307) and for inviting a revised, expanded version. We resubmit as a new manuscript implementing your central recommendation: to consolidate the OmniGene-4 and BioPAWS work into a single, balanced study with a fair benchmarking framework.

You were not fully convinced the paper offered a fair benchmarking framework, “particularly given that BioPAWS was also developed by your group.” We took this seriously and addressed it head-on. We have turned the benchmark into a first-class, openly released contribution — **BioPAWS-2**, a chat-form suite of 9 task families / 22 tasks (~306K examples), released on Hugging Face ([dnagpt/biopaws-2](https://huggingface.co/dnagpt/biopaws-2)) with all code on GitHub, so any group can score any model on it. Crucially, we then used our own benchmark to argue *against* a self-serving conclusion: a matched raw-Gemma-base control under identical joint fine-tuning shows our model’s biological continued-pretraining confers *no material advantage under supervised fine-tuning* (macro 0.769 vs. 0.760) — the opposite of self-benchmarking. OmniGene-4’s value is instead in deployment cost (~1.5 GPU-days, 4-bit GGUF on a single 24 GB GPU), modality breadth, and zero-shot remote-homology detection. We also release an MMseqs2 leakage audit and re-split every sequence task entity-disjoint (homology leakage 0.78→0.08); re-running all supervised results on the clean splits moves the headline by only −1.4 points, showing the leaderboard is not a memorization artifact.

The empirical core of the original submission (router-level CPT-vs-SFT decomposition, remote-homology results, modality-invariant transfer) is retained in full. All artefacts — dataset, model checkpoints, and training/evaluation/leakage-audit code — are public (github.com/maris205/omnigene4), so any leaderboard number is independently reproducible. A preliminary version is on bioRxiv (DOI 10.1101/2026.01.03.697478). The author declares no competing interests; GPU compute was donated by io.net, which exercised no influence over the research.

We are grateful for the path you offered toward a more balanced paper and welcome any further feedback.

Sincerely,

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